BILLIEBOT — MASTER WIRING & POWER SCHEMATIC

SHEET 1 OF 1 · REV A · per Build & Test Manual rev A

⚠ BENCH PSU (0.5 A limit) FOR FIRST POWER-UP — NEVER THE LiPo · LABEL EVERY WIRE WITH ITS CONNECTION ID

APPENDIX A · MASTER CONNECTION CHECKLIST (optional items D10 / D16 / P12 omitted from this build)

	ID	Connection	Method
☐	P1	Battery (+) → 20 A main fuse	XT60 / solder
☐	P3	Switch → +12 V bus strip	ring terminal
☐	P5	Bus [10 A] → L298N VMOT / GND	screw terminals
☐	P7	Bus [5 A] → GeeekPi PD board VIN	screw terminals
☐	P9	Bus [5 A] → GS2678 buck IN	screw terminals
☐	P11	5 V rail → MAX98357A VIN / GND	Dupont / solder
☐	P14	≥1000 µF 25 V cap across Jetson feed	solder, mind polarity
☐	D2	OAK-D Lite → Jetson blue USB 3.0 port	USB 3.0 C→A
☐	D4	Encoders L/R → Nano D3/D2 · D4/D7	Dupont, quadrature
☐	D6	L298N OUT1/2 · OUT3/4 → motors L / R	screw terminals
☐	D8	MLX90640 → Pi pins 1/3/5/6	I²C, 3.3 V
☐	D11	ReSpeaker → mini hub → Pi USB-A	USB 2.0 audio
☐	D13	Amp → 3 W 4 Ω speaker	2-wire screw term
☐	D15	Pi 5 ↔ GL-SFT1200 router	Wi-Fi 2.4 GHz

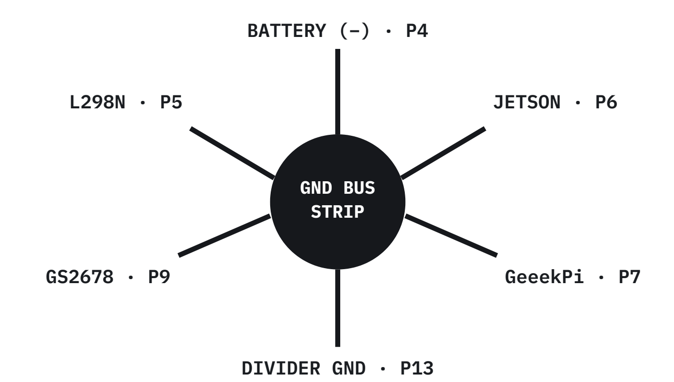
	ID	Connection	Method
☐	P2	Main fuse → master switch	spade
☐	P4	Battery (–) → GND bus strip (star point)	ring terminal
☐	P6	Bus [10 A] → Jetson barrel 5.5×2.5 mm ctr+	barrel plug
☐	P8	GeeekPi USB-C → Pi 5 USB-C	USB-C PD 5 V/5 A
☐	P10	GS2678 OUT → 5 V accessory strip (5.10 V)	screw terminals
☐	P13	Bus → 10k/3.3k divider → Nano A7	22 AWG
☐	D1	RPLidar A1 (UART-USB adapter) → Jetson	USB 2.0
☐	D3	Arduino Nano → Jetson (serial + Nano 5 V)	USB 2.0 A→Mini-B
☐	D5	Nano D6/D10/D9/D5 → L298N IN1–IN4	Dupont, 5 V PWM
☐	D7	BNO055 → Nano A4/A5 + 5 V/GND	I²C, 5 V
☐	D9	Pi Camera 3 NoIR → Pi CAM/DISP-0	22-pin FFC, CSI-2
☐	D12	MAX98357A → Pi pins 12/35/40	I²S
☐	D14	Jetson ↔ GL-SFT1200 router	Wi-Fi 2.4 GHz

FUSE & CURRENT BUDGET

BRANCH	FUSE	EST. MAX LOAD	WIRE
Main feed (P1)	20 A	≤10 A worst-case	16 AWG
Motors (P5)	10 A	~4 A both stalled	16–18 AWG
Jetson (P6)	10 A	1–2 A typ · 3 A pk	18 AWG
GeeekPi PD → Pi (P7)	5 A	~2.7 A @ 12 V in	18 AWG
GS2678 5 V rail (P9)	5 A	~2.6 A @ 12 V in	18 AWG

Whole-robot idle ≈ 1.5–2.5 A @ 12 V (Stage 4). Every fuse sits at the **bus end** of its branch. Motor (P5) and compute (P6–P8) wiring are separate home-runs to the strips — never daisy-chained. Pack: Venom 3S LiPo, 9.9–12.6 V, 44 Wh.

★ STAR-GROUND TOPOLOGY



All returns land **only** at the GND bus strip — one spoke per load, no daisy-chains, no second meeting point. Logic grounds (encoders, BNO055, I²C/I²S) ride inside their signal bundles and reference their host board, which itself returns to the star.

LEGEND

+12 V power
GND / return
+5 V / 5.1 V rail
battery-sense analog
USB cable
I²C (SDA dk / SCL lt)
I²S audio
FFC ribbon
PWM / DIR logic
encoder A phase
encoder B phase
encoder / logic 5 V
Wi-Fi link

STROKE = WIRE GAUGE
16 AWG
18 AWG
22 AWG
24 AWG / signal

SYMBOLS
fuse (rating beside)
dot = junction
no dot = crossing

P- / D-numbers in **chips** are the manual's connection IDs (distinct from Nano pin names). Highlighted pinout rows = pins used in this build.

BILLIEBOT

Apartment patrol robot — MVP configuration

TITLE	MASTER WIRING & POWER SCHEMATIC
DOC REF	Build & Test Manual rev A · §5–8, App. A
REV	A
DATE	2026-07-12
SHEET	1 / 1 · NTS
DRAWN	Claude — Sr. EE

Deviation from manual §5: L298N onboard-5V jumper **removed**; logic 5 V fed from Nano 5 V pin (per build pin map). PWM on IN1–IN4; ENA/ENB jumpers ON.